1. **Scenario:** You are working on a project that involves analyzing a dataset containing information about houses in a neighborhood. The dataset is stored in a CSV file, and you have imported it into a NumPy array named house_data. Each row of the array represents a house, and the columns contain various features such as the number of bedrooms, square footage, and sale price.
   Question: Using NumPy arrays and operations, how would you find the average sale price of houses with more than four bedrooms in the neighborhood?

House_ID	Bedrooms	Sqft	Sale_Price
H1	3	1400	7500000
H2	5	2200	12500000
H3	4	1800	9800000
H4	6	3000	18500000
H5	2	1100	5200000

2. **Scenario:** An email service provider wants to automatically classify incoming emails as spam or non-spam.Each email is represented using features such as number of links, number of capitalized words, and total email length. Historical email data is labeled as spam (1) or not spam (0).
   Question:
   Develop a Logistic Regression model using the given features.
   Predict whether a newly received email is spam or not.

Email_ID	Num_Links	Num_Caps_Words	Email_Length	Spam
E1	1	3	120	0
E2	6	15	300	1
E3	0	1	80	0
E4	4	10	250	1
E5	2	5	160	0

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4. **Scenario:** Linear Regression for Car Resale Price PredictionYou are a data analyst at a used-car marketplace. You want to predict the resale price of a car based on features such as car age (years), total kilometers driven, and engine capacity (cc). You have a dataset containing past car listings with these features and their final selling prices.
   Question: Write a Python program that loads the dataset containing Age, Kms_Driven, Engine_CC, and Resale_Price, trains a Linear Regression model using scikit-learn, accepts user input for a new car, and predicts the estimated resale price.

Car_ID	Age (Years)	Kms_Driven	Engine_CC	Resale_Price
C1	2	25000	1200	650000
C2	5	60000	1500	420000
C3	3	40000	1300	520000
C4	7	90000	1800	300000
C5	1	15000	1000	720000

5. **Scenario:** Loan Approval (CART Classification): A bank wants to decide whether to approve or reject a loan application. The decision depends on features like income, credit score, and existing loans. Past applications are labeled as Approved (1) or Rejected (0).
   Question: Using CART, build a Decision Tree Classifier for loan approval.
   Predict the class for a new applicant and display the decision rules.

Applicant_ID	Income (₹/Month)	Credit_Score	Existing_Loans	Approved
A1	45000	720	0	1
A2	30000	650	1	0
A3	60000	780	0	1
A4	28000	620	2	0
A5	52000	710	1	1